Basic Principles of Medicine 1
Module : Foundations to Medicine
FTM Lecture No: #22

Lecture Title

Genetics and Genomic Cases
(flipped classroom)

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St Georges University ©

Pre-Lecture - Required Reading and Objectives

- Refresh information found within the two lectures:
  - Molecular biology techniques and associated readings for that lecture
  - Use of Genomics in Diagnosis and associated readings for that lecture

- Objectives

SOM.MKIII.BPM1.1.FTM.3.GNET.0081	Analyze examples of molecular diagnosis obtained from genetic and genomic laboratory analysis

Lecture Objectives

SOM.MKIII.BPM1.1.FTM.3.GNET.0081	Analyze examples of molecular diagnosis obtained from genetic and genomic laboratory analysis

All these children have an extra chromosome 21
Family features can influence the phenotype

Ekanem Ekure, S.J. Patil, Girisha K.M., Antonio Richieri-Costa, and Vorasuk Shotelersuk.

A 1-year-old boy is brought to the physician for examination because of facial dysmorphology and developmental delay. Cytogenomic microarray analysis is suggested which includes both SNP analysis and array CGH. Which of the following best describes an aspect of this genomic test?

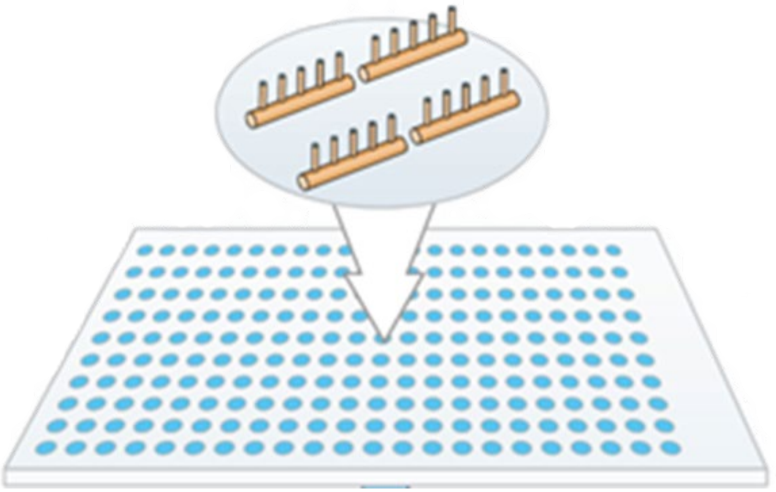
A. Uses combinations of PCR primers — 0%

B. Uses many immobilized probes — 0%

C. Relies on dideoxy chain termination — 0%

D. Arranges PCR fragments by size — 0%

E. Hybridizes ssDNA sequences to metaphase chromosomes — 0%



A 7-year-old boy with dysmorphology and intellectual impairment is brought to the physician for follow-up examination following return of test results. Previous testing includes array CGH and CGG repeat sizing of the *FMR1* gene; both of which were normal. Sequencing of all the protein coding regions of this boy's genome by NGS showed that he had a mutation in the *PQBP1* gene and Renpenning syndrome is diagnosed. Which of the following best describes the test used to diagnose this patient?

A. Whole genome sequencing	0%
✓ B. Whole exome sequencing	0%
C. Targeted gene sequencing	0%
D. Sanger sequencing	0%
E. ^{32}P labeled dideoxy sequencing	0%
F. Specific oligonucleotide directed sequencing	0%

A large family with several individuals affected with an AD disorder is being studied with PCR RFLP to identify a genomic locus associated with the disorder. An RFLP marker is found that is tightly associated with the phenotype. This allowed the researchers to identify a 10 million base pair region that is likely to have the gene with the causative mutation. Which of the following best describes this approach to diagnosis?

A. Comparative genome hybridization	0%
B. Linkage analysis	0%
C. Sizing of PCR products	0%
D. In situ hybridization	0%
E. Southern	0%

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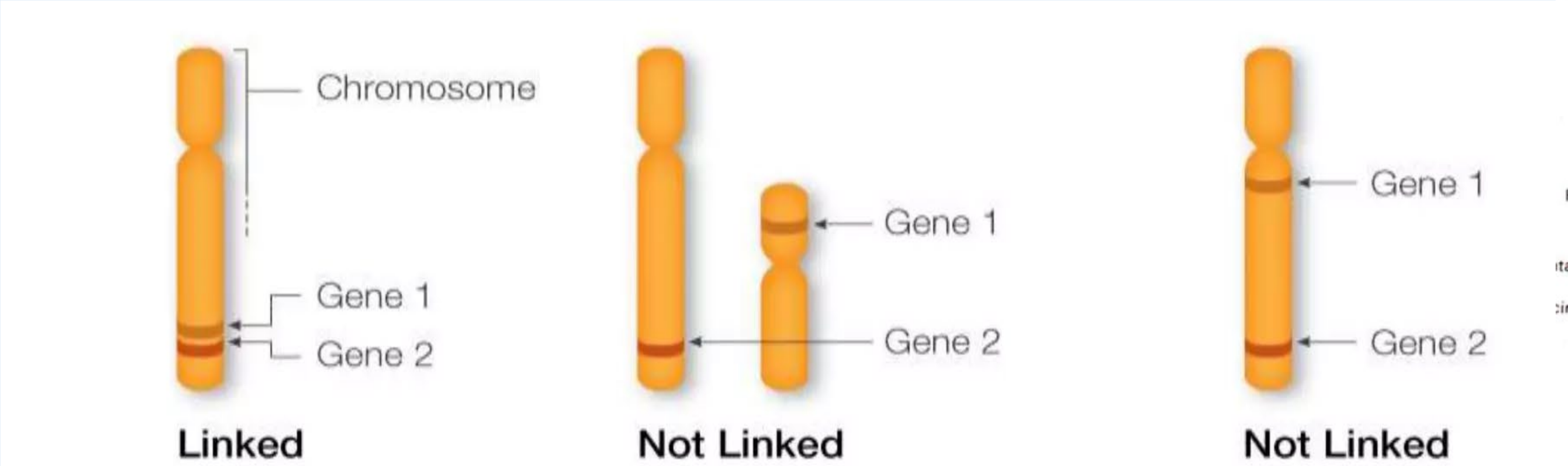
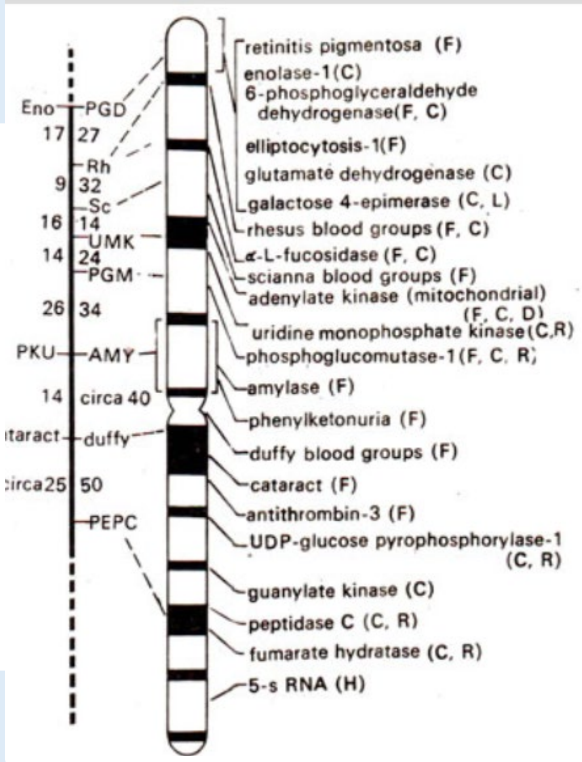
When a marker and a gene are so close to each other:

-There is a low probability that recombination happens between
-Another way to say it is there is a high probability that they will segregate together





1983 map of chromosome 1 using linkage

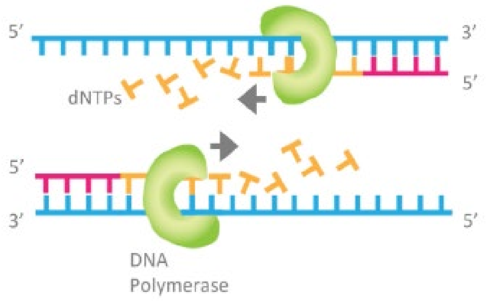
A large family with several individuals affected with an AD disorder is being studied with PCR RFLP to identify a genomic locus associated with the disorder. An RFLP marker is found that is tightly associated with the phenotype. This allowed the researchers to identify a 10 million base pair region that is likely to have the gene with the causative mutation. Which of the following best explains how the properties of DNA allow this study to be carried out?

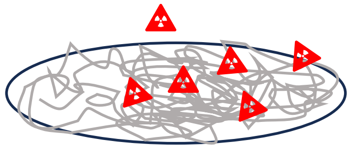
A. Telomeric DNA is found on chromosomal ends 0%
B. Restriction enzymes cut at palindromic sequences 0%
C. Introns are spliced from exons during RNA processing 0%
D. All DNA point mutations affect restriction sites 0%
E. Most human genes exhibit alternative splicing 0%
F. DNA is replicated by a semi-conservative mechanism 0%

5'-CAGCTG
5'-CAGCTG

A researcher synthesizes an 18 base deoxyribonucleic acid polymer. Conditions are created that allow this reagent to hybridize specifically, and only to, its exact match of single stranded DNA. Which of the following best describes this small polymer of single stranded DNA? (2 answers are correct)

A.	Restriction enzyme	0%
B.	PCR Primer	0%
C.	FISH probe	0%
D.	cDNA	0%
E.	Allele specific oligonucleotide	0%





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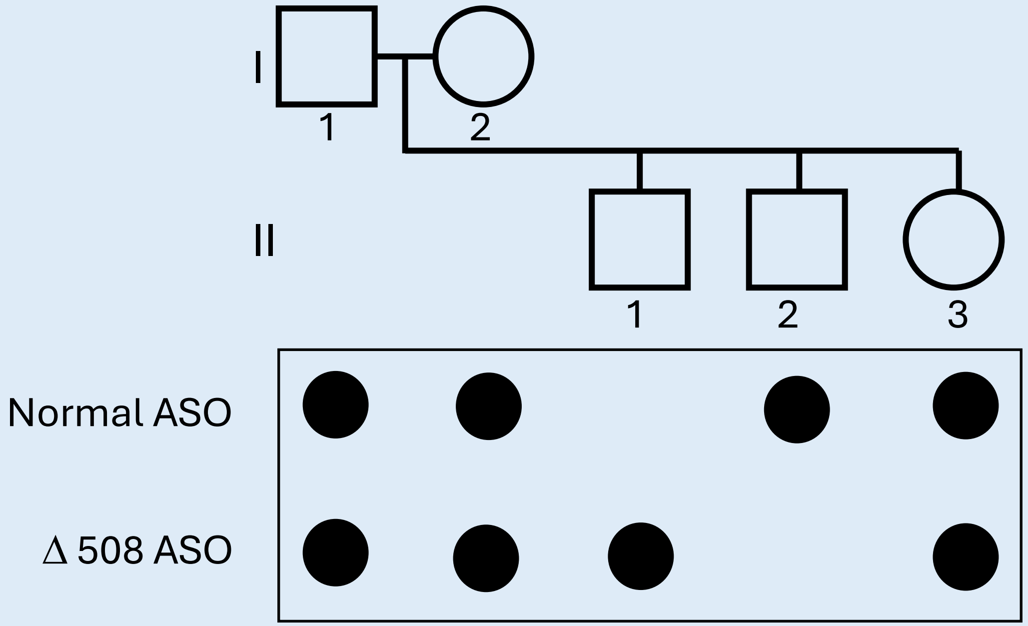
Vignette for the next 4 clicker questions: consider the ASO dot blot shown below which provides genotype for a family about the ΔF508 allele of the *CFTR* gene.

Cystic Fibrosis allele Δ508 has 3 bp deletion (AGA)

ASO for DNA sequence specific to Δ508 variant 5′-CACCAAAGATGATATTTTC-3′

ASO for normal DNA sequence 5′-CACCAATGATATTTTC-3′



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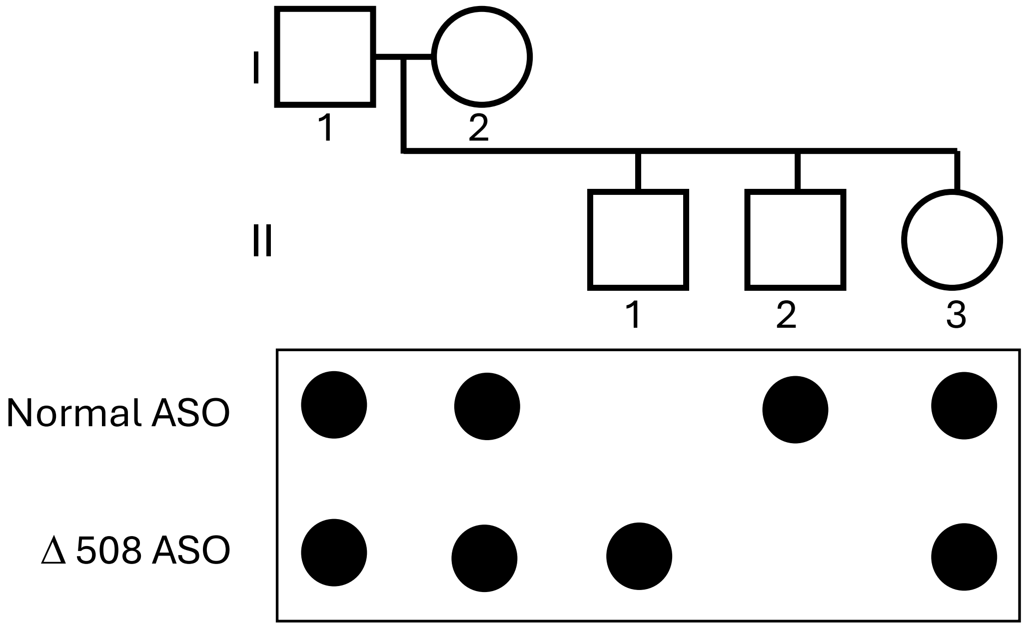
Which of the following best describes what the normal ASO probe will detect in this test?

A. The normal *CFTR* gene 0%

B. A normal genome 0%

C. The exact match of the 19 base sequence that is complementary to this oligonucleotide 0%



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Which of the following individuals in the pedigree and corresponding ASO test most likely has CF?

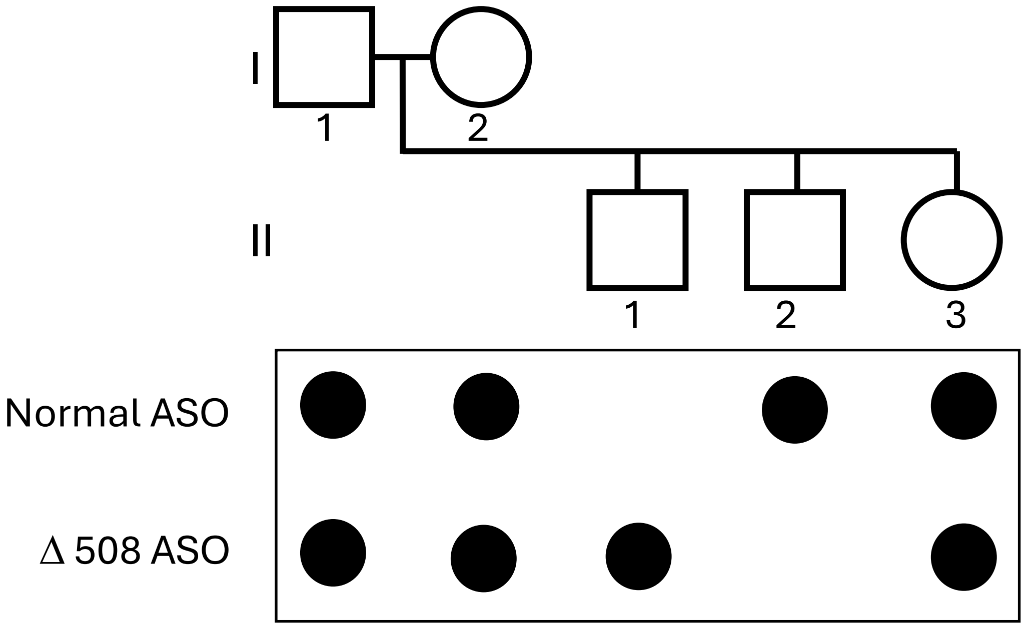
A. A 0%

B. B 0%

C. C 0%

D. D 0%

E. E 0%



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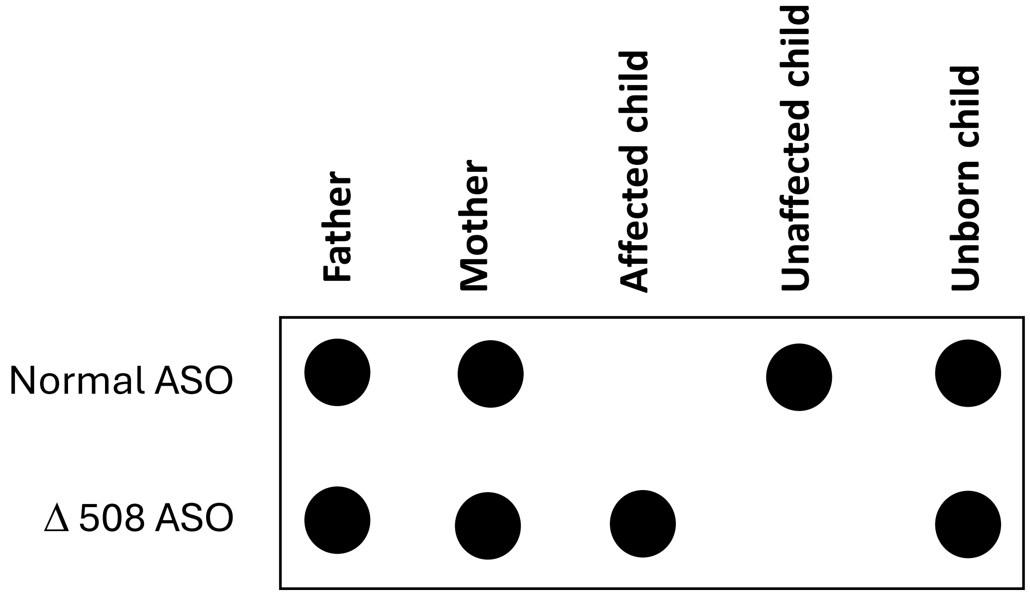
An ASO test for cystic fibrosis of the family is performed. The results of the test are depicted. The prenatal test for the unborn child is also performed. Which of the following is the best risk estimate that the unborn child will be affected?

A.	100%	0%
B.	50%	0%
C.	25%	0%
D.	10%	0%
E.	Nil	0%



Remember: If the mutation was NOT ΔF508, it would not be detected by ASO for ΔF508.

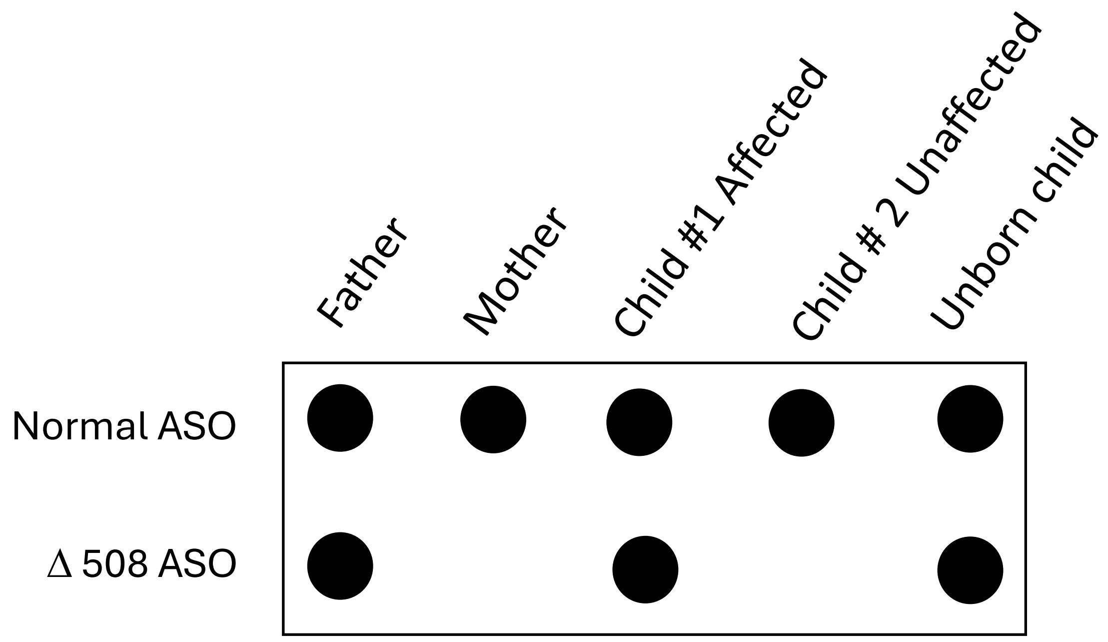
(In the second half of this course we will learn to calculate that only about 50% of CF patients are homozygous for the ΔF508)

The previous question: is the simple example

- We have genotyped each parent.
- Each parent is heterozygous for the ***same*** disease allele (the delF508 allele) of *CFTR*
- A child with CF who is born from these parents will be homozygous for dF508 allele.

This is a different family

Assume that parentage is <u>authentic</u> in the family tested. The first child of this family has CF. What would most likely be the mother's status?



A. Homozygous for normal *CFTR* alleles 25%

B. Heterozygous for *CFTR* disease alleles 25%

C. Homozygous for *CFTR* disease alleles 25%

D. We may not interpret this data to make a conclusion for the mother 25%

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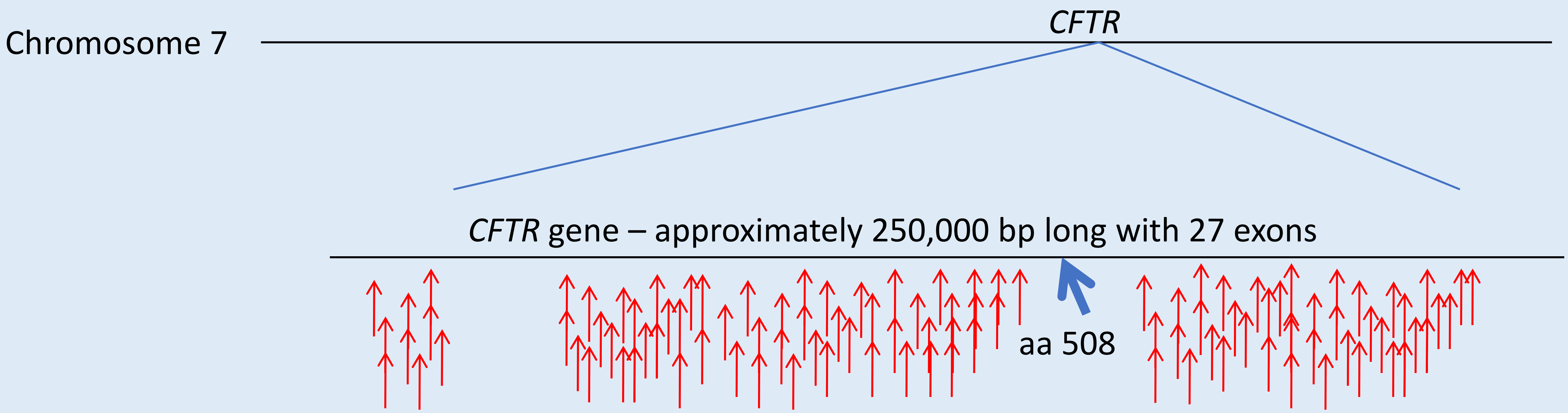
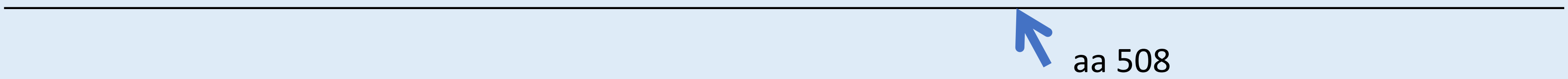
Explanation of the previous question:



The mother's DNA is probed with a wild type oligo and a Δ508 oligo, and she is found to be normal at the 508 locus. However, she had a child with the disease, and her husband is verified to be a carrier of the d508 mutation



Therefore, the mother must be a carrier of a mutation at some other critical region of her *CFTR* gene. Of course, this will not be detected by an ASO for the 15 bp region around the f508 locus.

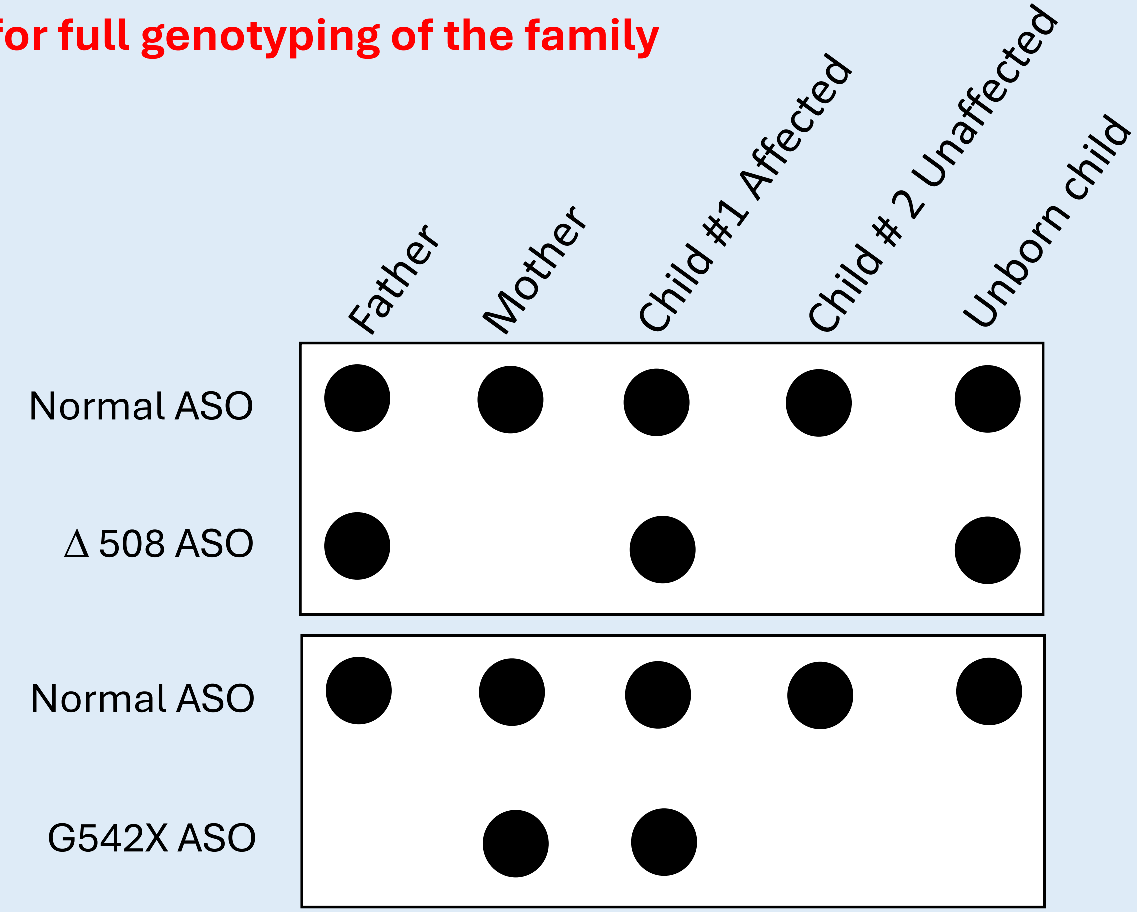
See problem set for the step-by-step tutorial about this application

This *CFTR* ASO blot example illustrates the importance of understanding the molecular distinction between allelic heterogeneity, and locus heterogeneity

As we discussed in the patterns of inheritance section, the individual who has an autosomal recessive disorder due to inheritance of two *different* loss of function variants is called ….

Compound heterozygote

This is data for full genotyping of the family



When the rest of the data is shown, it should become more clear

Now we see:

1. The mother is a carrier in a mutation other than ΔF508 (G542X)
2. Child #1 is affected as a compound heterozygote
3. Child #2 is homozygous for the normal allele
4. Child #3 (currently an unborn child) will be a carrier of ΔF508

A cohort of men who had their prostate surgically removed participated in the following study: First, they allowed researchers to isolate mRNA from their resected tumor and also sections of normal prostate tissue. Next, gene expression was analyzed. Which of the following tests was most likely used to compare this collection of expressed sequences to each other?

A.	SNP chip	0%
B.	microarray CGH	0%
C.	ASO blotting	0%
✓ D.	cDNA microarray	0%
E.	PCR RFLP	0%
F.	FISH	0%

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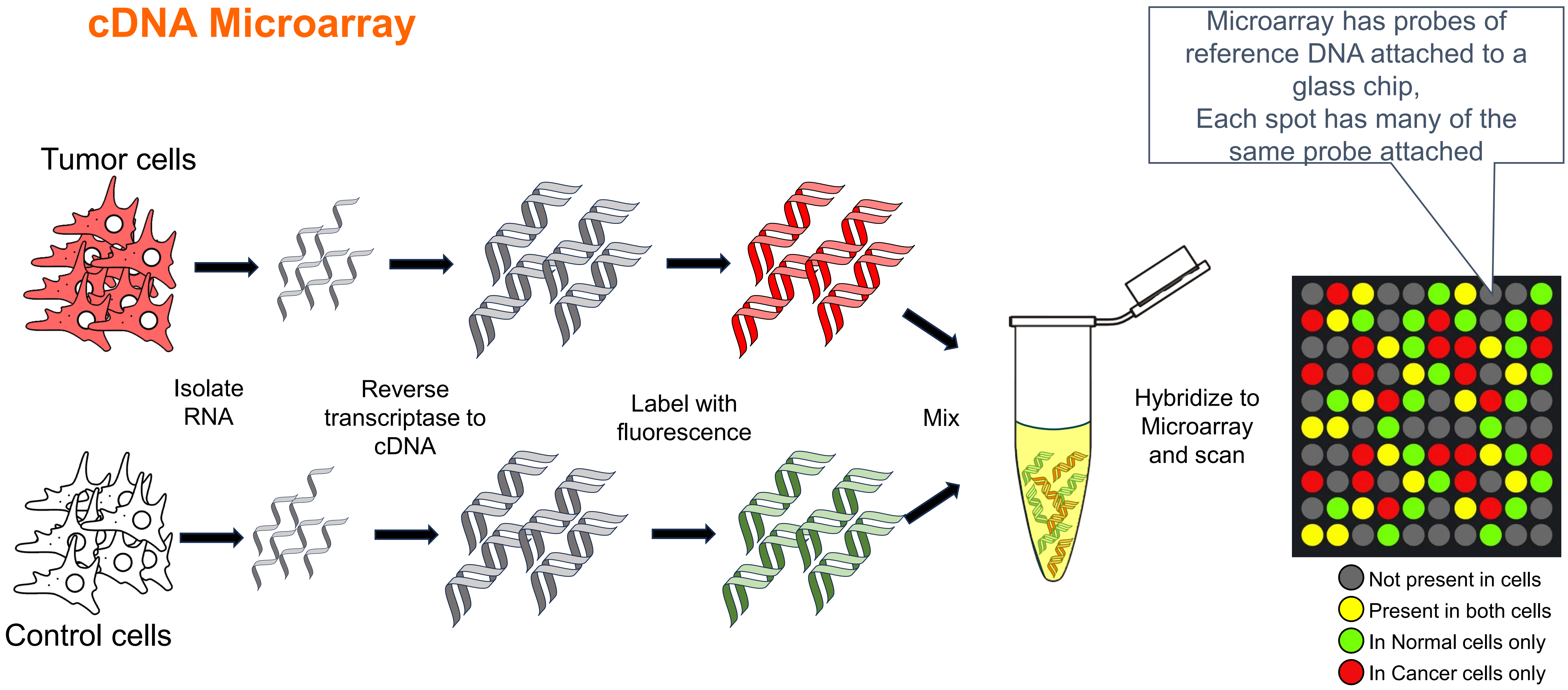
cDNA Microarray



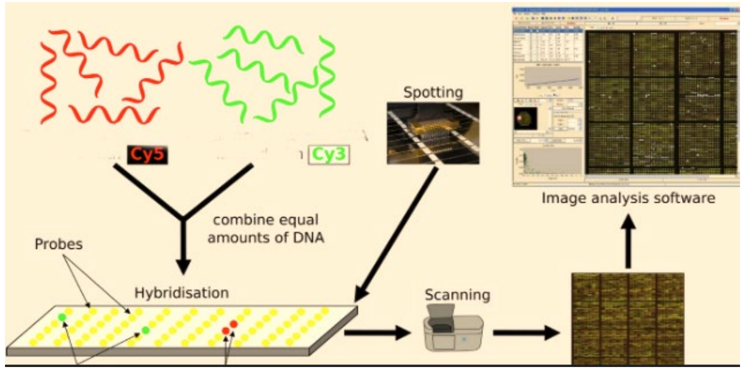
RNA sequencing with NGS technology

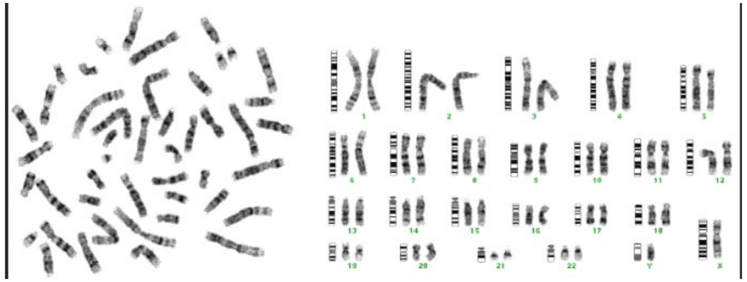
- RNA sequencing can be done with next-generation sequencing methods
  - **Called RNA Seq**

- DNA can be sequenced (easy) and mRNA may be sequenced as well
  - Can collect all transcripts from a collection of cells (from tissue biopsy, or culture)
    - Isolate mRNA and sequence it
    - Can tell relative abundance
      - Will get more sequence reads from mRNA that is more abundant
    - Can verify and/or reveal splice site mutations

A 13-month-old girl is brought to the physician because of developmental delay, motor delay, and dysmorphology. Which of the following is a first-tier test that should be given for this patient during the search for a genetic diagnosis? (2 answers are correct)

✔	A. Array CGH	0%
	B. RFLP by Southern blot	0%
	C. SKY FISH	0%
✔	D. G-banding	0%
	E. PCR sizing of CAG repeats	0%





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A 13-month-old girl is brought to the physician because of developmental delay, motor delay, and dysmorphology. Which of the following is a first-tier test that should be given for this patient during the search for a genetic diagnosis?

Notice that exome sequencing was not an option on the previous question…

As the cost of analysis of all the data collected by NGS decreases, (and the storage of the data decreases) exome, or genome sequencing is fast becoming a first-tier test (*or the primary test; in fact, one might argue that it is currently the first-tier test*) for individuals with neurodevelopmental delay.

A patient is suspected to have a genetic cause underlying his symptoms. A genetically controlled malformation syndrome is suspected. Karyotype of metaphase stage cells appears normal. A test is given that allows the entire genome to be analyzed and a deletion that is commonly associated with the phenotype is detected. Which of the following tests was most likely used?

✓ A.	Microarray CGH	0%
B.	FISH	0%
C.	cDNA array	0%
D.	G-banding	0%
E.	PCR RFLP	0%
F.	Spectral karyotype	0%
G.	Western blot	0%
H.	Measuring sizes of PCR products	0%

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Which of the following best fits the definition of a “genomic test”?

A. FISH for 22q11.2 deletion (DiGeorge) 0%
B. ASO blot panel of CFTR mutations 0%
C. PCR RFLP and linkage analysis 0%
D. Metaphase G-band karyotype 0%
E. Northern blot 0%

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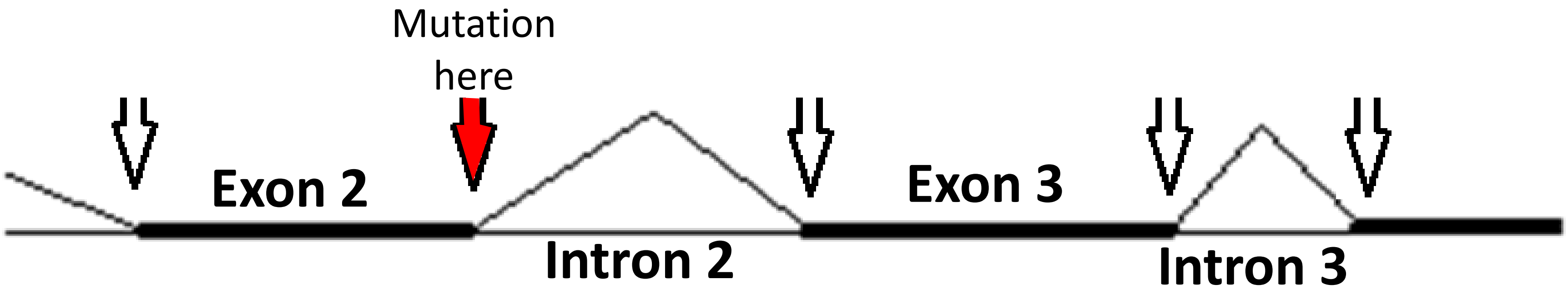
Genomic means that the entire genome is analyzed

- In a sense, G-band karyotype gives an overview of the entire genome
  - Problem is that G-band is of relatively low resolution – meaning that relatively small deletions or duplications will not be revealed
  - As a very rough guideline, the change must be at least 5 Mb or bigger
  - Better guideline, the change must affect the banding pattern, but this will be different for different places in the genome.
- FISH, ASO, RFLP, Southern, Northern, all look at one gene at a time

DNA sequencing shows that a patient with neurofibromatosis has a predicted pathogenic splice site mutation. Which of the following would be the best approach to verify that this splice site mutation actually has an effect on expression of the *NF1* gene?



A.	PCR amplify the gene	0%
B.	Sequence the mRNA	0%
C.	Do an ASO blot	0%
D.	Perform Southern blotting	0%
E.	Sequence the intron (DNA)	0%
F.	Use the microarray CGH	0%

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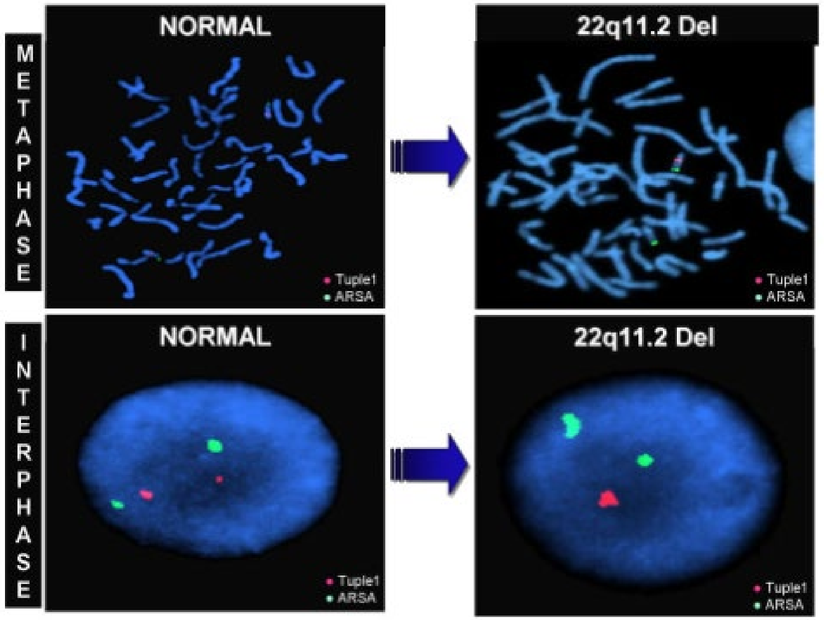
The karyotype for a 2-year-old boy with developmental delay and mild intellectual impairment appears normal. Using the microscope slides that were already prepared for the karyotype, it was shown that the boy actually has 22q11.2 ds (DiGeorge) after a single probe was used to identify the centromeres of both chromosome 22, and that one of these had a deletion in the "critical region". Which of the following tests was done to obtain the diagnosis in this patient?

A. ASO blot	0%
B. Southern	0%
C. Array CGH	0%
D. SNP Chip	0%
✔ E. FISH	0%



A 39-year-old woman comes to the physician because of muscle weakness and a reduced ability to release her grip after squeezing. For instance, she has difficulty releasing a doorknob after opening the door. A genetic test shows that she has an aberration causative of a genetic disorder. Diagnosis was done by showing that a PCR product amplified from her DNA was larger than would be expected in a person without the disorder. Which of the following best describes the pathogenic variant in this patient?

A. Triplet repeat expansion	0%
B. Point mutation	0%
C. Deletion of an exon	0%
D. Chromosomal rearrangement	0%
E. Loss of a chromosome	0%
F. Linkage analysis	0%

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A 2-year-old boy with intellectual disability is tested with various genetic tests. Standard karyotype appears normal. Array CGH detects an approximate 0.5 Mb deletion. Which of the following best explains the relatively low resolution of standard karyotype (G-banding) compared to array CGH?

✓	A. Each band is composed of many bp	0%
	B. Each band must be labeled with a different stain	0%
	C. Each chromosome has only one band	0%
	D. Interphase chromosomes are dispersed	0%
	E. It is not linked to a marker	0%

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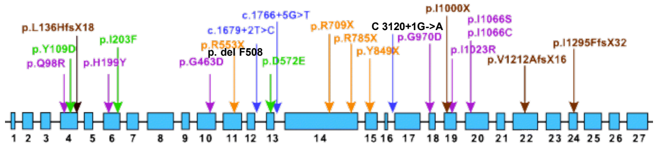
Link this concept: Allelic vs. Locus Heterogeneity

Allelic Heterogeneity	**Locus Heterogeneity**
Same gene	Different genes
Different mutations on same gene	Different genes have mutations leading to same syndrome
Example: Cystic Fibrosis: AR inheritance	Example: Treacher Collins: AR or AD inheritance
CFTR gene: Common mutations shown 	Mutations leading to non-functional proteins TCOF1 on chromosome 5 POLR1C on chromosome 6 POLR1D on chromosome 13

***CFTR* Mutation F508**	**Severe phenotype**	**TCOF1**	**Treacher Collins – AD**
***CFTR* Mutation p.G970D** ***CFTR* Mutation c.3120 +1G>A** ***CFTR* Mutation p.R117H** **……..**	**Moderate-severe phenotype** **Moderate-severe phenotype** **Mild phenotype**	**POLR1D** **POLR1C**	**Treacher Collins – AD** **Treacher Collins – AR**

Summary of microarray CGH (from DLA and Lecture):

Array CGH allows unbiased interrogation of the genome for the detection of submicroscopic deletions and duplication – these are called 'copy number variants' or CNV

- Array CGH is constructed by arranging ~0.5 - 1 million probes on a chip
  - Probes are immobilized on an array, hence it is called an array
- Each probe answers the following question:
  - Does the patient have more than, less than, or the same amount of DNA as would be expected to be seen in a 'normal' control reference genome.
- Do the math: The haploid genome is ~3 billion bp in size, so it ~300,000 probes are used
  - Then the average space between probes is ~10,000 bp (or 10 kb).
- So array CGH can detect deletions and duplications that involve approximately 10 kb.
  - Typically, the claim is 30,000 bp, so range is in the ten thousands of base pairs.
  - Remember, the limit for detection on a karyotype is something like 5 Mb (for change in banding patter) and this is seen with the microscope; so, array CGH is an enormous increase in resolution
  - Therefor array CGH can detect CNV that are undetectable by microscope (karyotype)

Summary of NGS (from DLA and Lecture):

Next-Gen Sequencing (NGS): obtained via massively parallel attainment of sequence reads

1. The patient’s DNA is chopped into small pieces
2. Can choose to pull out all DNA, only the exons, a group of related genes, or a single gene
3. The pieces that are pulled out are then linked with a common linker
4. The linked genomic fragments are stuck on to a surface
5. Allows all sequencing reactions to be initiated from a common primer
6. Many millions of sequence reads (sequencing reactions) can be obtained
7. This is considered a multiplex approach on a grand/mega scale
8. All of the sequence reads are relatively short (very short sequence reads)
9. These are all mapped to the previously known genomic sequence
10. Without the map, the sequence reads would be much less useful
    - Maybe close to useless
11. Analysis of variants – search for variants that are pathogenic
12. Advent in computer power makes all of this possible

More summary descriptions of genomic tests (from DLA and Lecture): Students must know some aspects of the tests used for genomic analysis

- SKY FISH does not detect deletions or duplications, it only paints each chromosome a specific color
- Western blot detects protein levels by binding an antibody to the protein which had been separated by polyacrylamide gel
- Biochemical analysis of enzyme activity are useful to detect heterozygous carriers of enzyme deficiency (such as Tay Sachs)
- G-banding will only detect very large deletions / duplications, we use as a rule of thumb about 5 Mb. If smaller than 5 Mb, they may be below detection limit for detection – recall discussion about 22q11.2 (DiGeorge syndrome)
- Array CGH is an unbiased test that is able to interrogate the genome to look for deletions and duplications
- FISH does allow unambiguous identification of a deletion (because only one of the chromosomes would light up for that area) from a metaphase spread of the chromosomes.
- The problem with FISH is that the appropriate probe would have to be chosen, as FISH only allows one test to be done at a time – the choice of doing the test is therefore ‘hypothesis driven’.
- FISH would not allow an estimation of the size of the deletion only that the size of the deletion is bigger than the FISH probe –
- FISH might not pick up a duplication because the area lighting up might be directly next to each other and will likely not be resolved (it would just appear as a single blob of light; this would depend on the deletion and the FISH probe used - so it’s technical).

Compare and Contrast:

- What is the difference between a “genetic” test, and a “genomic” test?

Summary of triplet repeat expansion testing:

In almost all cases diagnosis of triplet repeat expansion disorders employs PCR amplification

- PCR amplification using primers outside of the repetitive region
  - Primers are set to hybridize in the unique outside of the repeats
- Separation of the PCR products
  - Column chromatography (primary method because it gives more exact measurement and is more easily automated)
  - Agarose gel electrophoresis (the old method)
- Next Generation Sequencing typically cannot identify triplet repeat amplifications
  - Because repeats by their definition are repetitive – so they cannot be reliably mapped to the genome
  - Because of this each triplet repeat expansion disorder needs its own PCR / separation test
- In very rare cases, very long (super long), repetitive regions cannot be amplified by PCR (because the region is just too long).
  - In this case, Southern blot is used
  - We don’t discuss this in this course

Office Hours

- Email mmaj@sgu.edu
- Can do open office hours by zoom as well